DP Biology – A4.1 (HL extension)

How new species arise

- What evidence supports evolution?
- What do analogous and homologous structures show about what organisms share and how they differ?

HL statements A4.1.8–11. Textbook pp. 151–155.
Image: the Grand Canyon, a geographic barrier. CC BY-SA 2.5.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A4.1.8–9	Compare allopatric and sympatric speciation, and explain adaptive radiation
2	A4.1.10–11	Explain barriers to hybridization, and abrupt speciation by hybridization and polyploidy

The big picture: slow and sudden speciation

Allopatric

Separated by place

Sympatric

Same place, separated otherwise

Polyploidy

A new species in one generation

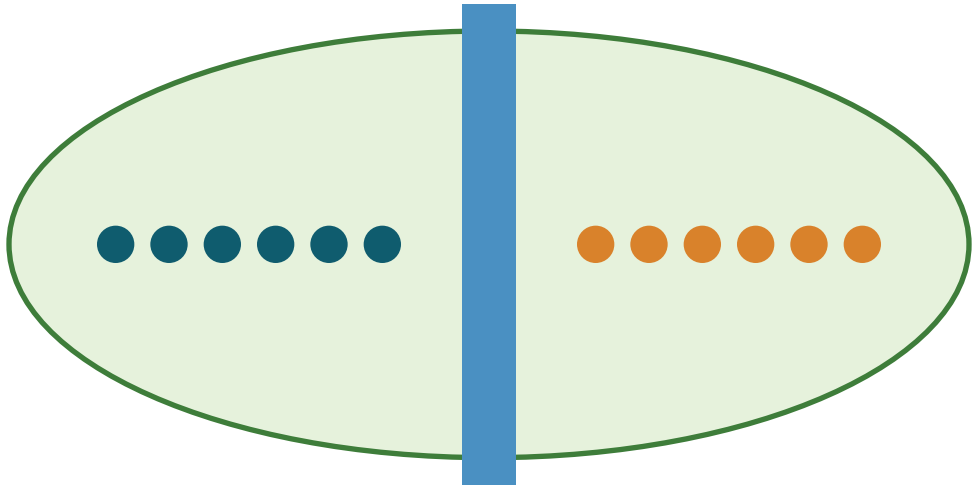
Every route to a new species starts with **reproductive isolation**: gene pools stop mixing, then diverge.

Big question

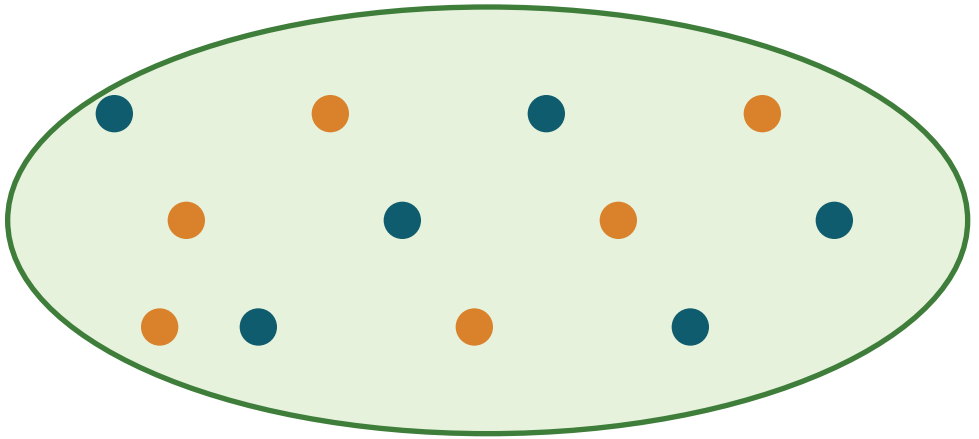
How can a new plant species appear in a single generation?

Allopatric and sympatric speciation

A4.1.8



Allopatric: a barrier divides them



Sympatric: same place, mixed, but not interbreeding

Same in both

Gene pools separate; selection and mutation make them diverge until they can no longer interbreed. The difference is how isolation arises.

Three kinds of reproductive isolation

A4.1.8



Western meadowlark. Kevin Cole, CC BY 2.0.

Type	Example
Geographic	Antelope squirrels either side of the Grand Canyon
Temporal	Spotted skunks: ranges overlap, breed in different seasons
Behavioural	Meadowlarks look alike but sing different songs

Try it

Which example leads to allopatric speciation, and which to sympatric?

Breeding at different times

A4.1.8

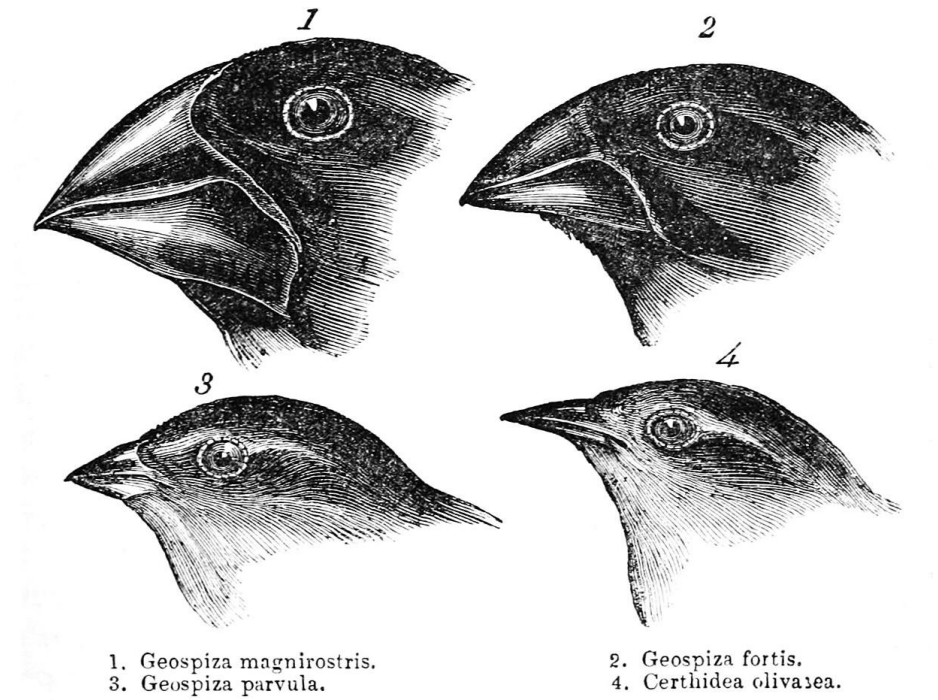


Key idea

Same place, different breeding seasons: no gene flow, so they evolve apart. Times shown are schematic.

Adaptive radiation

A4.1.9



Beaks of Galapagos finches, drawn in 1845. Public domain.

One ancestor diversifies into many species, each filling a different **niche**.

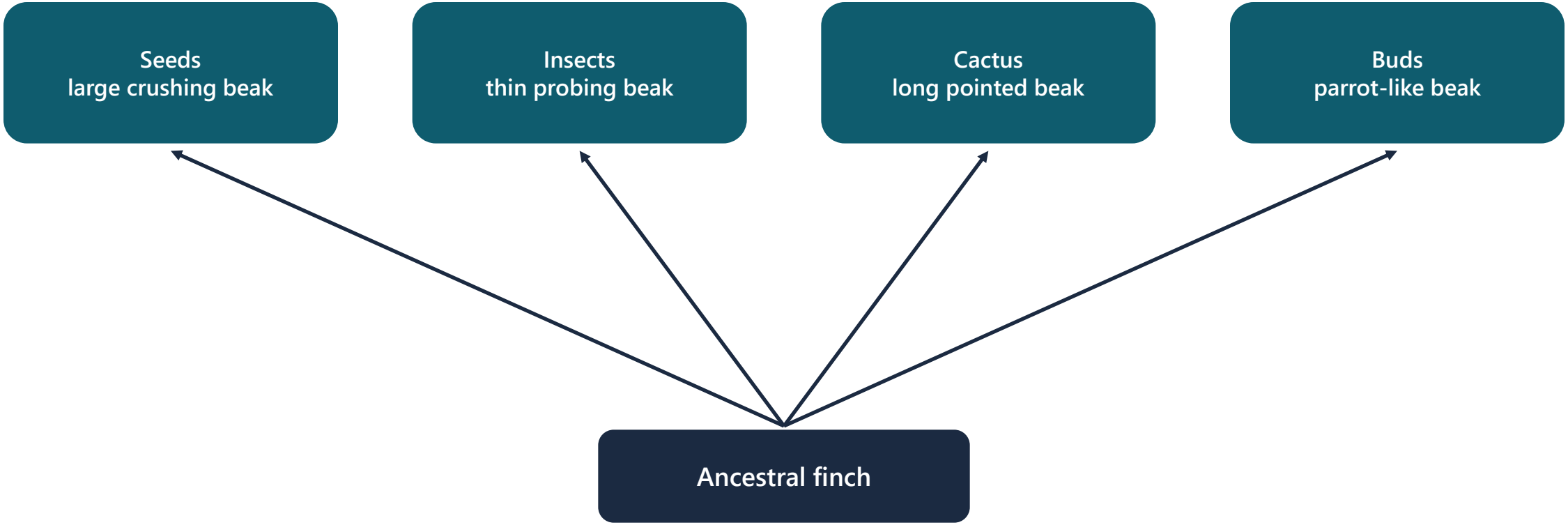
Vacant niches (new islands) give the opportunity. Different resources mean close relatives **coexist**.

So

Adaptive radiation increases biodiversity.

One ancestor, many niches

A4.1.9



Notice

Each species uses a different resource, so competition is reduced.

A plant example: bromeliads

A4.1.9



A bromeliad: leaves form a water-holding tank. Dinkum, CC0.

- Some root in bog or sand
- Some collect rain and debris in a leaf tank
- Some get nutrients from ants

Think

Why might competition push close relatives to use different resources?

Hundreds of species in one lake

A4.1.8–9

Lake Victoria holds over 500 cichlid species, most evolved in about 15 000 years.

Adaptive radiation

Jaws and teeth for algae, snails, insects or fish: different niches.

Kept apart by colour

Females choose males by colour: behavioural isolation in one lake (sympatric).

Discuss

Pollution made the water murky. What might happen to these species?

Check-in: class 1

A4.1.8–9

1. Distinguish allopatric and sympatric speciation.
2. Give three kinds of reproductive isolation.
3. Define adaptive radiation.
4. Explain how it raises biodiversity.
5. Give an example of sympatric speciation.

How to use this

Answer without your notes, then check against slides 3–10.

Keeping alleles apart

A4.1.10

Before mating

Species-specific courtship, songs, timing: individuals mate only with their own species

After mating

Hybrids die or are sterile: a mule (63 chromosomes) passes on no alleles

Why barriers are favoured

Sterile offspring waste resources, so selection favours individuals that avoid mating with other species. Barriers keep species distinct.

Why hybrids are sterile

A4.1.10–11



diploid: pairs



hybrid: no matching partners



doubled: every one paired

Key idea

Homologous chromosomes must pair in meiosis. Doubling the chromosomes of a hybrid gives each a partner again.

What is polyploidy?

A4.1.11

A **polyploid** has more than two sets of chromosomes. It can arise when meiosis or mitosis fails to separate chromosomes.

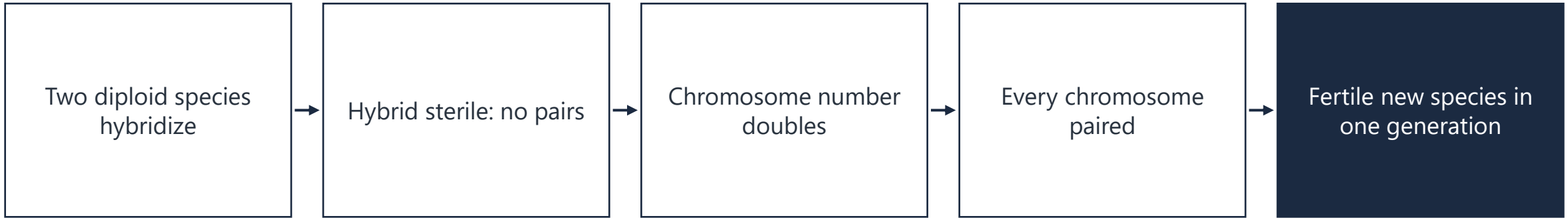
Type	Origin
Autopolyploid	Extra sets from one species
Allopolyploid	Sets from two species, after hybridization

Examples

Common in plants: wheat (hexaploid), kiwifruit, strawberries.

Abrupt speciation by polyploidy

A4.1.11



Why instant

Crossing back to a parent gives odd numbers of sets (triploid), which are sterile. So the polyploid is isolated at once.

Big question

A fertile new plant species can arise in one generation: this is speciation without geographic separation.

Knotweeds and smartweeds

A4.1.11



A smartweed, Persicaria maculosa. The Cosmonaut, CC BY-SA 2.5 ca.

The genus **Persicaria** has many species formed by hybridization and polyploidy; many are **allopolyploids**.

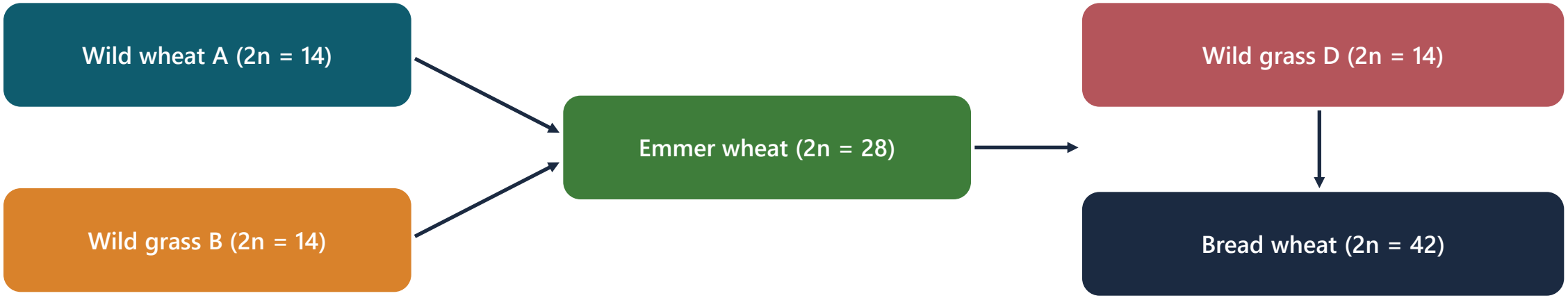
Polyploids can establish quickly in new habitats.

Try it

Two diploid parents each have 2n = 22. How many chromosomes in an allotetraploid?

Wheat: built by polyploidy

A4.1.11



Read it

Two rounds of hybridization and doubling gave bread wheat six sets of chromosomes (hexaploid). Simplified.

Working with data: chromosome numbers

A4.1.11

Chromosome numbers in related plants (simplified values for practice).

Plant	Chromosomes
P	22
Q	44
R	66
S	88

1. Find the basic number (x).
2. State the ploidy of each.
3. Why can Q and P not have fertile offspring?

Practice: barriers and polyploidy

A4.1.10–11

Explain

Why selection favours mating barriers.

Outline

The steps of speciation by polyploidy.

Calculate

2n of an allohexaploid from parents with × = 7.

Use

hybrid, sterile, polyploid, homologous, isolation

Check-in: class 2

A4.1.10–11

1. Give a pre-mating and a post-mating barrier.
2. Explain why barriers are favoured by selection.
3. Define polyploidy.
4. Explain how polyploidy makes a species in one generation.
5. Distinguish autopolyploid and allopolyploid.

How to use this

Answer without your notes, then check against slides 12–19.

Common mistakes to avoid

Not this	But this
Sympatric speciation needs a physical barrier.	It happens without one.
Polyploidy is always harmful.	Many crops and wild plants are polyploid.
Speciation always takes millions of years.	Polyploidy can do it in one generation.

Summary: how new species arise

A4.1.8–11

	Key idea
A4.1.8	Allopatric vs sympatric; types of isolation
A4.1.9	Adaptive radiation
A4.1.10	Barriers to hybridization
A4.1.11	Abrupt speciation by polyploidy

Check that you can...

1. **Compare** allopatric and sympatric speciation
2. **Outline** types of isolation
3. **Explain** adaptive radiation
4. **Explain** abrupt speciation by polyploidy

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between allopatric and sympatric speciation.
2. Explain how adaptive radiation lets close relatives live in the same area.
3. Explain why a new polyploid plant is reproductively isolated from its parents.

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